

ASSIGNMENT # 03

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Q1. Given Relations

- **COUNTRY** (Country_ID, Country_Name, Region, GDP_Rank)
- **THREAT** (Threat_ID, Threat_Name, Category, Severity_Level)
- **INCIDENT** (Incident_ID, Threat_ID, Country_ID, Date_Reported, Impact_Score)
- **RESPONSE_TEAM** (Team_ID, Country_ID, Team_Name, Specialization)
- **ACTION_TAKEN** (Incident_ID, Team_ID, Action_Type, Resolution_Status)

Queries and Relational Algebra Expressions

1. Countries that reported critical-level threats in 2025

$$\pi_{\text{Country_Name}} \left(\sigma_{\text{Severity_Level} = \text{"Critical"} \wedge \text{Year}(\text{Date_Reported}) = 2025} \left(((\text{INCIDENT} \bowtie_{\text{INCIDENT.Threat_ID} = \text{THREAT.Threat_ID}} \text{THREAT}) \bowtie_{\text{INCIDENT.Country_ID} = \text{COUNTRY.Country_ID}} \text{COUNTRY}) \right) \right)$$

Explanation: Selects countries that reported critical threats in 2025.

2. Response teams that resolved ransomware-related incidents

$$\pi_{\text{Team_Name}} \left(\sigma_{\text{Threat_Name} = \text{"Ransomware"} \wedge \text{Resolution_Status} = \text{"Resolved"}} \left((((\text{THREAT} \bowtie_{\text{THREAT.Threat_ID} = \text{INCIDENT.Threat_ID}} \text{INCIDENT}) \bowtie_{\text{INCIDENT.Incident_ID} = \text{ACTION_TAKEN.Incident_ID}} \text{ACTION_TAKEN}) \bowtie_{\text{ACTION_TAKEN.Team_ID} = \text{RESPONSE_TEAM.Team_ID}} \text{RESPONSE_TEAM}) \right) \right)$$

Explanation: Retrieves response teams that successfully resolved ransomware incidents.

3. Countries without any cyber response team

$$\pi_{\text{Country_Name}}(\text{COUNTRY}) - \pi_{\text{Country_Name}} \left(\text{COUNTRY} \bowtie_{\text{COUNTRY.Country_ID} = \text{RESPONSE_TEAM.Country_ID}} \text{RESPONSE_TEAM} \right)$$

Explanation: Finds countries with no record in the RESPONSE_TEAM relation.

4. Threats reported by more than one country

$$\pi_{\text{Threat_Name}} \left(\sigma_{\text{count}(\text{Country_ID}) > 1} \left(\gamma_{\text{Threat_ID}, \text{Threat_Name}; \text{count}(\text{Country_ID})} \right. \right. \\ \left. \left. (\text{THREAT} \bowtie_{\text{THREAT.Threat_ID} = \text{INCIDENT.Threat_ID}} \text{INCIDENT}) \right) \right)$$

Explanation: Groups incidents by threat and selects those reported by multiple countries.

5. AI-based attack incidents with impact score greater than 80

$$\pi_{\text{Incident_ID}, \text{Country_Name}, \text{Impact_Score}} \left(\sigma_{\text{Category} = \text{"AI Attack"} \wedge \text{Impact_Score} > 80} \right. \\ \left. ((\text{THREAT} \bowtie_{\text{THREAT.Threat_ID} = \text{INCIDENT.Threat_ID}} \text{INCIDENT}) \right. \\ \left. \bowtie_{\text{INCIDENT.Country_ID} = \text{COUNTRY.Country_ID}} \text{COUNTRY}) \right)$$

Explanation: Displays AI-based attacks with impact score above 80 and their countries.

Question #2: Relational Algebra and SQL Queries

Database Schema and Query Interpretations

Given Schema:

Cafe(cafeID, cafeName, city, managerID)
Employee(empID, name, address, gender, salary, cafeID)
Customer(custID, name, address, phone)
Orders(orderID, custID, cafeID, totalAmount, orderDate)
MenuItem(itemID, itemName, category, price, isAvailable)
OrderDetails(orderID, itemID, quantity)
Supplier(supplierID, name, city, contactNo)
Supplies(supplierID, itemID, supplyDate, cost)

1. Unavailable Items Priced Above 500

$$\pi_{\text{itemName}} (\sigma_{\text{isAvailable}=0 \wedge \text{price} > 500} (\text{MenuItem}))$$

Meaning: Retrieve all menu item names that are unavailable and priced above 500.

SQL:

```
SELECT itemName
FROM MenuItem
WHERE isAvailable = 0 AND price > 500;
```

2. Beverages Over 300 or Desserts

$$\pi_{itemName}(\sigma_{(category='Beverage' \wedge price > 300) \vee category='Dessert'}(MenuItem))$$

Meaning: List all items that are either beverages costing more than 300 or desserts.

SQL:

```
SELECT itemName
FROM MenuItem
WHERE (category = 'Beverage' AND price > 300)
      OR category = 'Dessert';
```

3. Available Snack Items with Price

$$\pi_{itemName, price}(\sigma_{category='Snack' \wedge isAvailable=1}(MenuItem))$$

Meaning: Display item names and prices for all available snack items.

SQL:

```
SELECT itemName, price
FROM MenuItem
WHERE category = 'Snack' AND isAvailable = 1;
```

4. Available Snacks with Price Divided by 100

$$\pi_{itemName, \frac{price}{100}}(\sigma_{category='Snack' \wedge isAvailable=1}(MenuItem))$$

Meaning: Show available snacks with their prices divided by 100.

SQL:

```
SELECT itemName, price / 100 AS price_divided
FROM MenuItem
WHERE category = 'Snack' AND isAvailable = 1;
```

5. Suppliers Providing "Cappuccino"

$$\pi_{supplierID}(Supplies \bowtie MenuItem \bowtie \sigma_{itemName='Cappuccino'}(MenuItem))$$

Meaning: Find supplier IDs of suppliers who supplied the item named "Cappuccino".

SQL:

```
SELECT DISTINCT s.supplierID
FROM Supplies s
JOIN MenuItem m ON s.itemID = m.itemID
WHERE m.itemName = 'Cappuccino';
```

6. Employees Living in the Same City as Their Cafe

$$\pi_{\text{Employee.name, Cafe.cafeName}} \left(\sigma_{\text{Employee.cafeID} = \text{Cafe.cafeID} \wedge \text{Employee.address} = \text{Cafe.city}} (\text{Employee} \bowtie \text{Cafe}) \right)$$

Meaning: Retrieve names of employees and their cafe names where employee address matches cafe city.

SQL:

```
SELECT e.name, c.cafeName
FROM Employee e
JOIN Cafe c ON e.cafeID = c.cafeID
WHERE e.address = c.city;
```

7. Employees with Salary Not Equal to 50,000

$$\pi_{\text{empID}}(\sigma_{\text{salary} \neq 50000}(\text{Employee}))$$

Meaning: List employee IDs of all employees whose salary is not 50,000.

SQL:

```
SELECT empID
FROM Employee
WHERE salary <> 50000;
```

8. Suppliers Operating in Both Karachi and Lahore

$$\pi_{\text{supplierID}}(\sigma_{\text{city}='Karachi'}(\text{Supplier})) \cap \pi_{\text{supplierID}}(\sigma_{\text{city}='Lahore'}(\text{Supplier}))$$

Meaning: Find supplier IDs of suppliers who have branches or operations in both Karachi and Lahore.

SQL:

```
SELECT supplierID
FROM Supplier
WHERE city = 'Karachi'
INTERSECT
SELECT supplierID
FROM Supplier
WHERE city = 'Lahore';
```

9. Items Never Supplied by Any Supplier

$$\pi_{itemName}(MenuItem) - \pi_{itemName}(MenuItem \bowtie Supplies)$$

Meaning: List names of menu items that have never been supplied by any supplier.

SQL:

```
SELECT itemName
FROM MenuItem
WHERE itemID NOT IN (
    SELECT itemID
    FROM Supplies
);
```